Roll No. | | | | | | | |

End Semester Examination December 2025

Name of the Course: M.Tech Structural Engg.
Name of the Paper: Mathematical Modeling of structures
Time: 3:00 Hours

Semester: I
Course Code: MSE 105
Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each question are twenty.
(iv) Each question carries 10 marks

Q1	**(20marks)**	
(a)	Solve by Gauss elimination and backward sweep $\begin{bmatrix} 1 & -1 & 2 \\ 2 & -2 & 3 \\ 1 & 1 & 1 \end{bmatrix} x = \begin{Bmatrix} -8 \\ -20 \\ -2 \end{Bmatrix}$	
(b)	Solve the given equation using Gauss Seidel iteration scheme $4x - y - z = 3$, $-2x + 6y + z = 9$, $-x + y + 7z = -6$	**CO1**
(c)	Solve the system of equations $\begin{bmatrix} 2 & 1 & 1 & -2 \\ 4 & 0 & 2 & 1 \\ 3 & 2 & 2 & 0 \\ 1 & 3 & 2 & -1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} -10 \\ 8 \\ 7 \\ -5 \end{Bmatrix}$ using the Gauss elimination method with partial pivoting.	
Q2	**(20 marks)**	
(a)	Find the values of y when (i) $x = 160$ ft. using Newton's forward interpolation formula (ii) $x = 410$ ft. using Newton's backward interpolation formula	**CO2**

x (height in ft.)	y (distance in ft.)
100	10.63
150	13.03
200	15.04
250	16.81
300	18.42
350	19.90
400	21.27

(b) Find the polynomial f (x) by using Lagrange's formula and hence find f (3) for

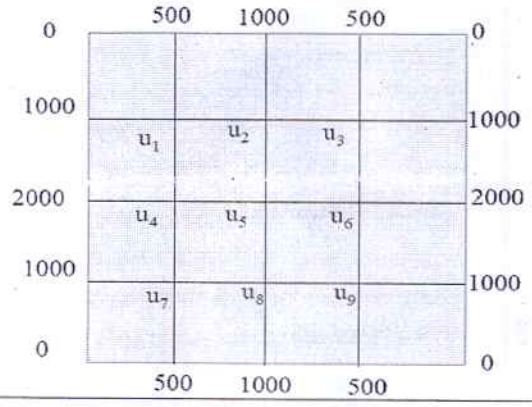
x	0	1	2	5
f(x)	2	3	12	147

(c)

Q3 **(20 marks)** **CO3**

(a) What is a hypothesis test? What are Type 1 and Type 2 errors?

(b) Solve the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for the following square mesh with boundary condition.



(c) What do you understand by covariance? What are its types?

Q4 **(20 marks)** **CO4**

(a) Explain the purpose and principle of the Chi-Square test. Describe two common applications and why it is called a "non-parametric test."

(b) A die is rolled 60 times. Observed counts: 1→8, 2→9, 3→10, 4→12, 5→11, 6→10. Find out the value of Chi-square.

(c) The following data relate X and Y:

X	1	2	3	4	5
Y	2	4	5	4	5

Find the Karl Pearson correlation coefficient.

Q5 **(20 marks)** **CO5**

(a) A shop records 100 customers classified into three product choices: A→30, B→50, C→20. Find Chi-square if choices are equally likely (so expected each = 33.33).

(b) Write all the steps involved in mathematical modelling of structures?

(c) The following data give the rainfall (X, mm) and corresponding crop yield (Y, quintals/ha):

Rainfall (X)	800	850	900	950	1000
Yield (Y)	30	32	35	37	38

Find the regression equation of Y on X.
Estimate yield when rainfall = 920 mm.